第二章 薛定谔方程

§1 薛定谔方程

§2 无限深势阱中的粒子

§3 势垒穿透

§4 谐振子

§1　薛定谔(Schrödinger)方程

一. 光子波动方程的建立和物理启示

1. 经典光矢量E满足的波动方程

$$\frac{1}{c^2}\frac{\partial^2 E(x,t)}{\partial t^2}=\frac{\partial^2 E(x,t)}{\partial x^2}$$

c为光速

2. 光子的波动方程

- 用光子的波函数 Ψ 代替E，得到方程

$$\frac{1}{c^2}\frac{\partial^2 \Psi(x,t)}{\partial t^2}=\frac{\partial^2 \Psi(x,t)}{\partial x^2}$$

光子的平面波函数

$$\Psi(x,t)=Ae^{\frac{i}{\hbar}(p_x x-\varepsilon t)}$$

代入得

$$(\frac{\varepsilon^2}{c^2})\Psi(x,t)=(p_x^2)\Psi(x,t) \qquad \frac{\varepsilon^2}{c^2}=p_x^2$$

光子的波动方程导致爱因斯坦能量关系

比较

$$\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\Psi(x,t)=\frac{\partial^2}{\partial x^2}\Psi(x,t)$$

$$\varepsilon^2 \sim \frac{\partial^2}{\partial t^2} \qquad p_x^2 \sim \frac{\partial^2}{\partial x^2}$$

$$\varepsilon \sim C_1\frac{\partial}{\partial t}, \quad p_x \sim C_2\frac{\partial}{\partial x}, \quad (C_1)^2=(C_2)^2$$

假定：能量和动量对应微分运算

$$\varepsilon \sim C_1 \frac{\partial}{\partial t}, \quad p_x \sim C_2 \frac{\partial}{\partial x}, \quad (C_1)^2 = (C_2)^2$$

爱因斯坦能量关系 $$\frac{\varepsilon^2}{c^2} = p_x^2$$

$$\frac{1}{c^2}\left(C_1 \frac{\partial}{\partial t}\right)^2 = \left(C_2 \frac{\partial}{\partial x}\right)^2$$

作用到波函数上
得到波动方程 $$\frac{1}{c^2}\frac{\partial^2 \Psi(x,t)}{\partial t^2} = \frac{\partial^2 \Psi(x,t)}{\partial x^2}$$

$$\Psi(x,t) = Ae^{\frac{i}{\hbar}(p_x x - \varepsilon t)}$$

$$\hat{\varepsilon}\Psi(x,t) \Rightarrow C_1 \frac{\partial}{\partial t} Ae^{\frac{i}{\hbar}(p_x x - \varepsilon t)} = -C_1 \frac{i\varepsilon}{\hbar} Ae^{\frac{i}{\hbar}(p_x x - \varepsilon t)}$$

如果希望 $\hat{\varepsilon}\Psi(x,t) = \varepsilon\Psi(x,t)$

$$-C_1 \frac{i}{\hbar} = 1 \qquad C_1 = i\hbar$$

同理 $\hat{p}_x \Psi(x,t) = p_x \Psi(x,t)$ $\qquad \hat{p}_x \Rightarrow C_2 \frac{\partial}{\partial x}$

$$C_2 \frac{i}{\hbar} = 1 \qquad C_2 = -i\hbar \qquad (C_1)^2 = (C_2)^2$$

二. 能量和动量的“算符”表示

1. “经典力学量”和“算符”的对应关系

假定 $\varepsilon \Rightarrow i\hbar\frac{\partial}{\partial t},\quad p_x \Rightarrow -i\hbar\frac{\partial}{\partial x}$

$x \Rightarrow x \qquad , i=\sqrt{-1}$

2. 算符

定义能量算符, 动量算符和坐标算符

$$\boxed{\hat{E} \equiv i\hbar\frac{\partial}{\partial t} \qquad \hat{p}_x \equiv -i\hbar\frac{\partial}{\partial x} \qquad \hat{x} \equiv x}$$

例1：能量、动量和坐标算符对沿x方向传播自由平面波波函数的作用

$$\Psi(x,t) = Ae^{\frac{i}{\hbar}(p_x x - Et)}$$

$$\hat{E}\Psi(x,t) = i\hbar\frac{\partial}{\partial t}\left(Ae^{\frac{i}{\hbar}(p_x - Et)}\right) = E\Psi(x,t)$$

$$\hat{p}_x\Psi(x,t) = -i\hbar\frac{\partial}{\partial x}\left(Ae^{\frac{i}{\hbar}(p_x - Et)}\right) = p_x\Psi(x,t)$$

$$\hat{x}\Psi(x,t) = x\Psi(x,t)$$

3. 建立光子的波动方程

- 利用对应关系得“算符关系等式”

$$\frac{E^2}{c^2} = p_x^{\ 2} \quad \Rightarrow \quad \frac{\hat{E}^2}{c^2} = \hat{p}_x^{\ 2}$$

- 把“算符关系等式”作用在波函数上得到光子的波动方程

三. 建立薛定谔方程

1. 薛定谔方程

描述一个质量为m的非相对论实物粒子在势场中的状态随时间的变化，

2. 建立薛定谔方程

经典 $$E=\frac{p_x^2}{2m}+U(x,t)$$

算符 $$\hat{E}=\frac{\hat{p}_x^2}{2m}+U(\hat{x},t)=\frac{\hat{p}_x^2}{2m}+U(x,t)$$

作用在波函数上 $$\hat{E}\Psi(x,t)=[\frac{\hat{p}_x^2}{2m}+U(\hat{x},t)]\Psi(x,t)$$

得 $$\boxed{i\hbar\frac{\partial}{\partial t}\Psi(x,t)=[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+U(x,t)]\Psi(x,t)}$$

三维情况：

$$\hat{\vec{p}} = \hat{p}_x \vec{i} + \hat{p}_y \vec{j} + \hat{p}_z \vec{k} = -i\hbar\nabla$$

$$\boxed{i\hbar \frac{\partial}{\partial t}\Psi(\vec{r},t) = [-\frac{\hbar^2}{2m}\nabla^2 + U(\vec{r},t)]\,\Psi(\vec{r},t)}$$

3. 哈密顿量（Hamiltonian）

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + U(\vec{r},t) = \frac{\hat{p}^2}{2m} + U(\vec{r},t)$$

用哈密顿量表示薛定谔方程

$$\boxed{i\hbar \frac{\partial}{\partial t}\Psi(\vec{r},t) = \hat{H}\Psi(\vec{r},t)}$$

四. 一维定态薛定谔方程

$$\hat{H}=\frac{\hat{p}_x^2}{2m}+U(x,t)$$

若 $\frac{\partial \hat{H}}{\partial t}=0$，或$U(x)$与时间无关，

则薛定谔方程可分离变量.

1. 分离变量

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+U(x)$$

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t)=\hat{H}\ \Psi(x,t)$$

设 $\Psi(x,t)\equiv\Phi(x)\,T(t)$

$$i\hbar\frac{dT(t)}{dt}\Phi(x)=[\hat{H}\Phi(x)]T(t)$$

$$i\hbar\frac{dT(t)}{dt}\frac{1}{T(t)}=\frac{1}{\Phi(x)}\hat{H}\Phi(x)\quad=E$$

$$\boxed{i\hbar\frac{dT(t)}{dt}=ET(t)\quad\sim（1）,}$$

$$\boxed{\hat{H}\Phi(x)=E\Phi(x)\quad\sim（2）}$$

2. 振动因子

方程（1）的解为 $T(t)=Ce^{-\frac{i}{\hbar}Et}$ —振动因子

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t)\ =E\ \Psi(x,t)$$

E代表粒子的能量

3. 一维定态薛定谔方程

$$\hat{H}\Phi(x) = E\Phi(x)$$

$$\boxed{[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)]\Phi(x) = E\Phi(x)}$$

4. 定态

能量取确定值的状态

定态波函数 $\Phi_E(x)$

$$\Psi_E(x,t) = \Phi_E(x)T(t) = C\Phi_E(x)e^{-\frac{i}{\hbar}Et}$$

定态波函数随时间演化，
几率密度不随时间改变

例2：一维自由粒子

$$U = 0 \qquad \hat{H} = \frac{\hat{p}_x^2}{2m} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

分离变量

$$\Psi_E(x,t) = C\Phi_E(x)\, e^{-\frac{i}{\hbar}Et}$$

$$\hat{H}\Phi(x) = E\Phi(x)$$

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\Phi(x) = E\Phi(x) \quad \rightarrow \quad \Phi''(x) + k^2\Phi(x) = 0$$

$$k^2 = \frac{2mE}{\hbar^2} = \frac{p_x^2}{\hbar^2}$$

$$\Phi(x) = A\cos kx + B\sin kx$$

$$\Phi(x) = Ae^{-ikx} + Be^{ikx}$$

$$\Psi_E(x,t) = Ae^{\frac{i}{\hbar}(-p_x x - Et)} + Be^{\frac{i}{\hbar}(p_x x - Et)}$$

一维自由粒子

$A = 0 \qquad \Psi_E(x,t) = Be^{\frac{i}{\hbar}(p_x x - Et)}$ ***x*** 正向平面波

$B = 0 \qquad \Psi_E(x,t) = Ae^{\frac{i}{\hbar}(-p_x x - Et)}$

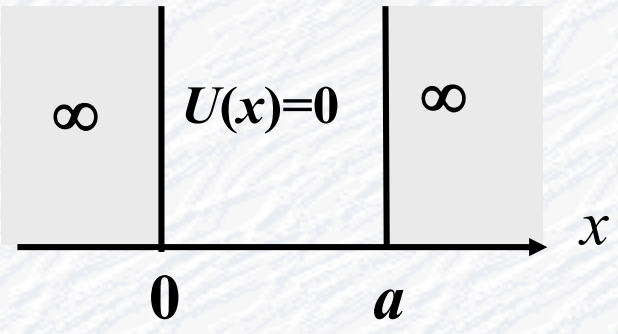
x 负向平面波

§2 无限深势阱中的粒子

1. 势函数

$$U(x)=0 \quad (0<x<a)$$

$$U(x)=\infty \quad (x\le 0 \text{或} x\ge a)$$



2. 哈密顿量

$$\hat{H}=-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+U(x)$$

3. 定态薛定谔方程

$$\Psi_E(x,t)=C\Phi_E(x)\,e^{-\frac{i}{\hbar}Et}$$

$$\hat{H}\Phi(x)=E\Phi(x)$$ 能量本征方程

- 阱内： $$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\Phi(x)=E\Phi(x)$$

令 $$k^2=\frac{2mE}{\hbar^2}$$ 则阱内方程为

$$\Phi''(x)+k^2\Phi(x)=0$$

- 阱外： $$[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\infty]\Phi(x)=E\Phi(x)$$

4. 分区求通解

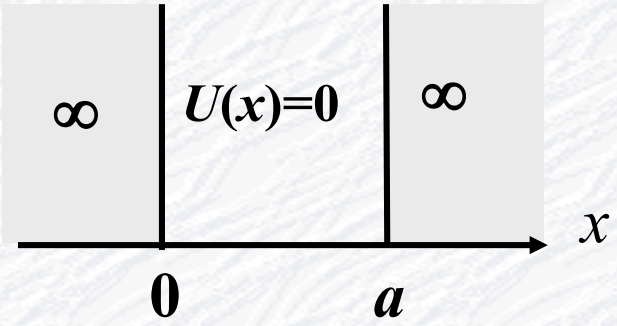
- 阱内： $$\Phi(x)=A\cos kx+B\sin kx$$

式中A和B是待定常数

- 阱外： $$\Phi(x)=0$$

5. 由波函数自然条件和边界条件定特解

$$\Phi(x) = A\cos kx + B\sin kx$$



$$\Phi(0) = 0 \;\Rightarrow\; A = 0$$

$$\Phi(a) = 0 \;\Rightarrow \sin ka = 0 \quad , \quad (B \neq 0)$$

$$ka = n\pi,\ (k \neq 0) \rightarrow k = \frac{n\pi}{a},\ n = 1,2,3,\cdots$$

$$\Phi(x) = B\sin kx$$

$$\Phi_n(x) = B\sin k_n x, \quad k_n = \frac{n\pi}{a}, \quad n = 1,2,3,\ldots$$

(1) 能量本征值 eigenvalue

$$k^2 = \frac{2mE}{\hbar^2} \qquad \therefore E_n = \frac{\pi^2\hbar^2 n^2}{2ma^2}, \qquad n = 1,2,3,\cdots$$

- 能量取分立值（能级） 能量量子化
- 当 $n \to \infty$ 时，量子化 -- 连续

$$\frac{E_{n+1} - E_n}{E_n} = \frac{2n+1}{n^2} \to 0$$

- 最低能量(零点能) $E_1 = \dfrac{\pi^2\hbar^2}{2ma^2} > 0$ — 波动性

只要有粒子, 就有能量

(2) 本征函数系 $\Phi_n(x) = B\sin k_n x$

$$\Phi_n(x) = \sqrt{\frac{2}{a}}\sin\frac{n\pi}{a}x \qquad (n = 1,2,3,\cdots)$$

(3) 本征函数系的正交归一性

$$\delta_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

可证 $$\int_0^a \Phi_m^*(x)\Phi_n(x)dx = \delta_{m,n}$$

(4) 几率密度

$$W_n(x) = |\Phi_n(x)|^2 = \frac{2}{a}\sin^2\frac{n\pi}{a}x$$

当 $n \to \infty$ 时，量子→ 经典

- **驻波 (Standing wave)**

$$\Phi_n(x)=\sqrt{\frac{2}{a}}\sin k_n x \qquad (n=1,2,3,\cdots)$$

$$k_n a = n\pi \qquad \lambda_n=\frac{2a}{n} \qquad p_n=\frac{nh}{2a}$$

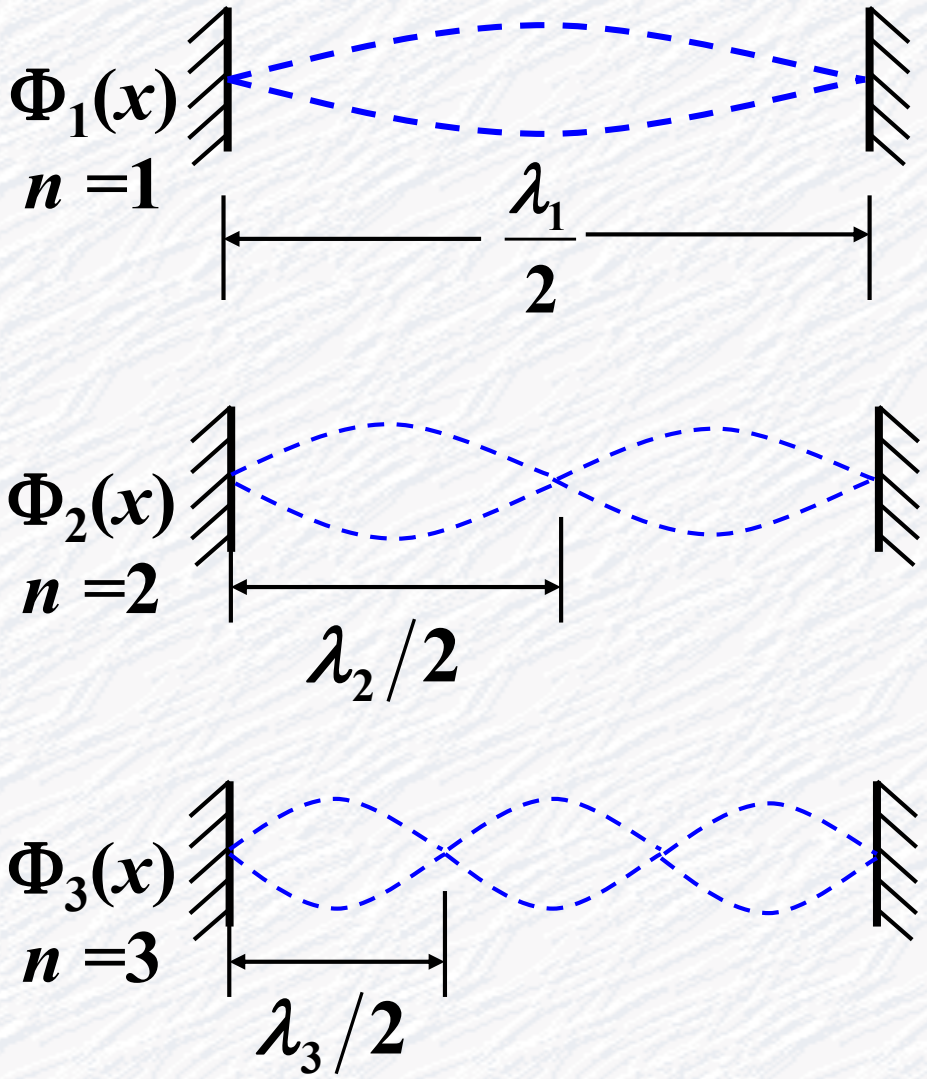
$$E_n=\frac{p_n^2}{2m}=\frac{n^2h^2}{8ma^2}=\frac{\pi^2\hbar^2n^2}{2ma^2}$$

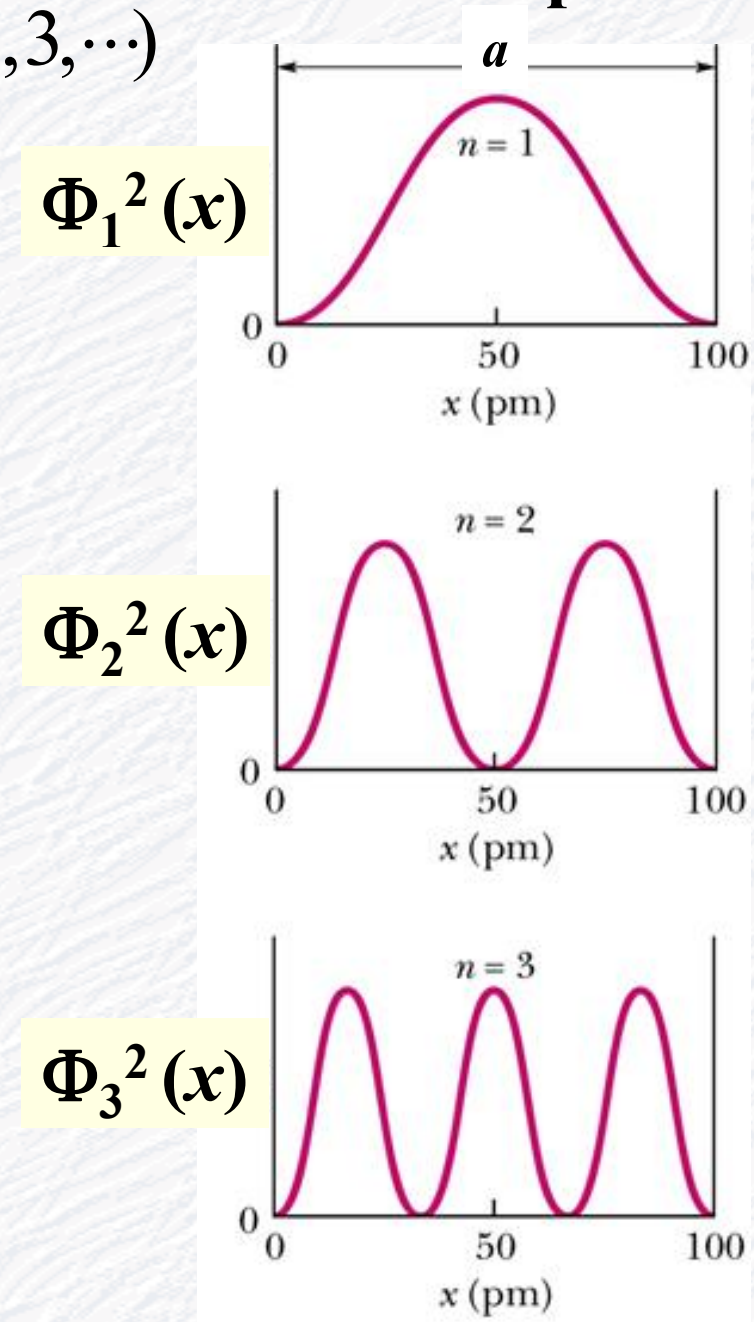
$$\Psi_n(x,t)=C_n\Phi_n(x)\,e^{-\frac{i}{\hbar}E_n t}$$

两端振幅为零的驻波

$$\Phi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{2\pi}{\lambda_n} x \qquad (n = 1, 2, 3, \cdots)$$

a = 100pm

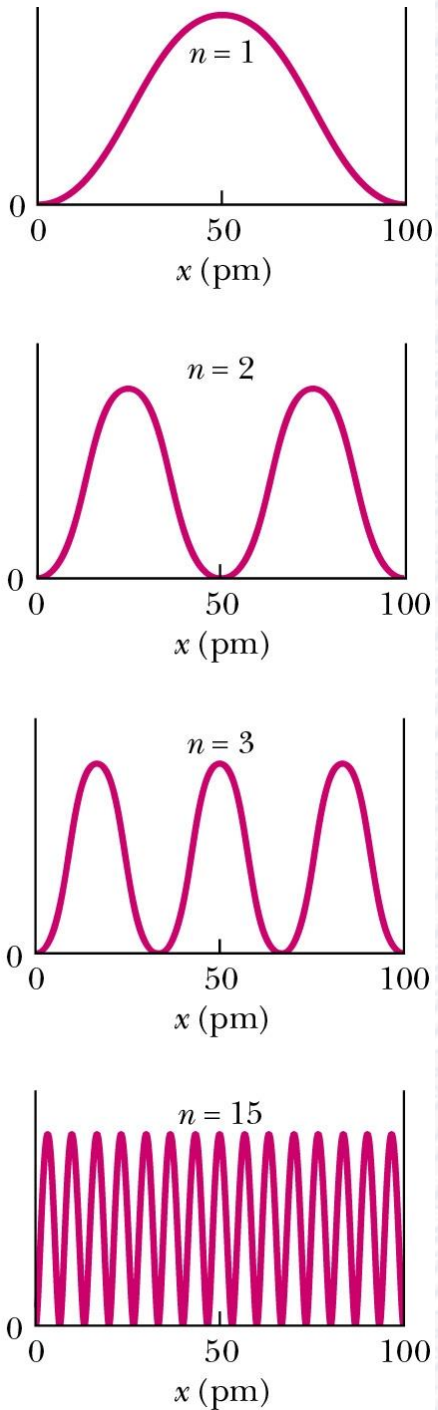


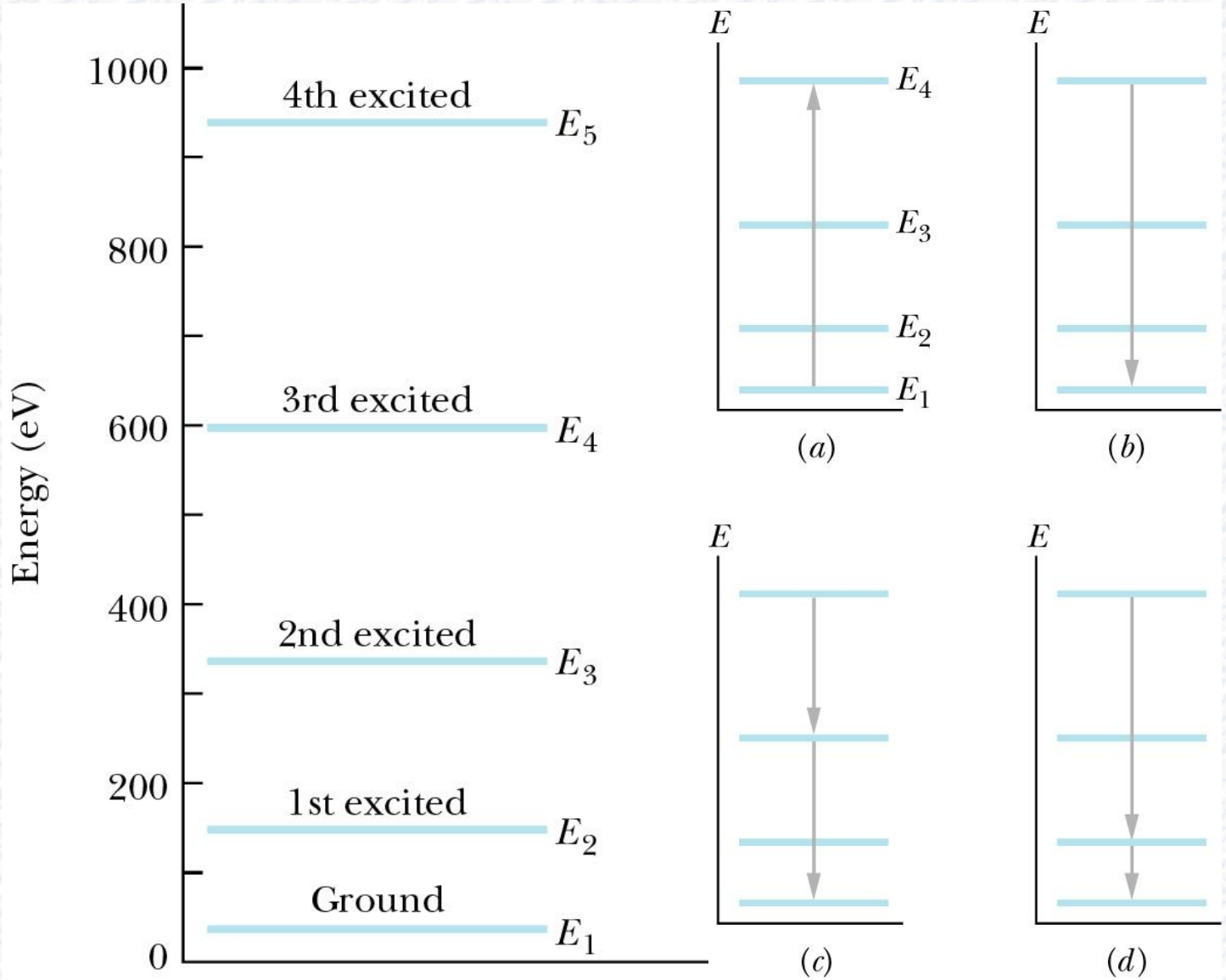


a = 100pm

能级

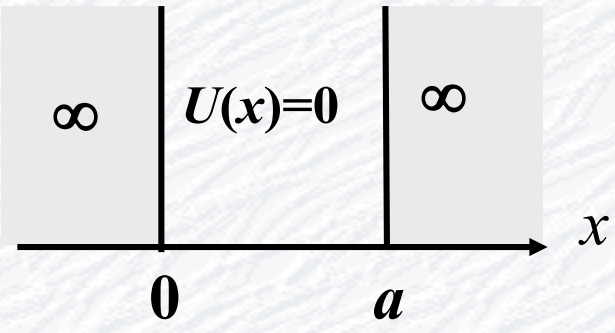
Excitation and de-excitation





$$i\hbar\frac{\partial}{\partial t}\Psi(x,t)=[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+U(x,t)]\Psi(x,t)$$

$$\Psi_E(x,t)=C\Phi_E(x)\,e^{-\frac{i}{\hbar}Et}$$



$$E_n=\frac{\pi^2\hbar^2n^2}{2ma^2},\qquad n=1,2,3,\cdots$$

$$\Phi_n(x)=\sqrt{\frac{2}{a}}\sin\frac{n\pi}{a}x$$

线性方程

$$\psi=c_1\psi_1+c_2\psi_2$$

通解

$$\boxed{\Psi(x,t)=\sum_{n=1}^{\infty}C_n\Phi_n(x)e^{-\frac{i}{\hbar}E_nt}}$$

C_n 初始条件和归一化条件确定

例3： 在阱宽为a 的无限深势阱中，一个粒子 $t = 0$ 时的状态为

$$f(x) \propto \sin\frac{\pi x}{a} - \sin\frac{2\pi x}{a}$$

问：1）粒子 t 时刻的波函数

2）可能测到的能量值和相应几率？

3）粒子某时在 $< a/2$ 区域出现的几率？

解：1）$t = 0$ 波函数归一化

$$\Psi(x,0) = Cf(x) = C(\sin\frac{\pi x}{a} - \sin\frac{2\pi x}{a})$$

已知无限深势阱中粒子的

$$\Phi_n(x)=\sqrt{\frac{2}{a}}\sin\frac{n\pi}{a}x,\qquad E_n=\frac{\pi^2\hbar^2n^2}{2ma^2}\qquad n=1,2,3,\cdots$$

$$\Psi(x,0)=C[\Phi_1(x)-\Phi_2(x)]$$

$$\int_0^a\Psi^*(x,0)\Psi(x,0)dx=1$$

$$C^*C\int_0^a[\Phi_1^*(x)-\Phi_2^*(x)][\Phi_1(x)-\Phi_2(x)]dx=1$$

$$2C^*C=1\qquad C=\frac{1}{\sqrt{2}}$$

$$\Psi(x,0)=\frac{1}{\sqrt{2}}[\Phi_1(x)-\Phi_2(x)]$$

$$\Psi(x,t)=\sum_{n=1}^{\infty}C_n\Phi_n(x)e^{-\frac{i}{\hbar}E_n t}$$

$$C_1=\frac{1}{\sqrt{2}}\quad C_2=-\frac{1}{\sqrt{2}}\quad C_n=0\qquad n=3,4,5,\cdots$$

$$\Psi(x,t)=\frac{1}{\sqrt{2}}[\Phi_1(x)e^{-\frac{i}{\hbar}E_1 t}-\Phi_2(x)e^{-\frac{i}{\hbar}E_2 t}]$$

非定态的解

2）测量能量：可能测到的值只能是本征值

$$\Psi(x,t)=\frac{1}{\sqrt{2}}[\Phi_1(x)e^{-\frac{i}{\hbar}E_1t}-\Phi_2(x)e^{-\frac{i}{\hbar}E_2t}]$$

$$E_1=\frac{\pi^2\hbar^2}{2ma^2}1^2,\quad E_2=\frac{\pi^2\hbar^2}{2ma^2}2^2$$

*力学量 $\hat{A}$ 的平均值

$$\overline{A}=\int_{-\infty}^{+\infty}\Psi^*(x,t)\hat{A}\Psi(x,t)dx$$

能量平均值

$$\overline{E}=\int_{-\infty}^{+\infty}\Psi^*(x,t)\hat{E}\Psi(x,t)dx$$

$$\Psi(x,t) = \sum_{n=1}^{\infty} C_n \Phi_n(x) e^{-\frac{i}{\hbar} E_n t}$$

$$i\hbar \frac{\partial}{\partial t} \Psi(x,t) = \hat{H} \Psi(x,t) \qquad \hat{H} \Phi_n(x) = E_n \Phi_n(x)$$

$$\bar{E} = \int_{-\infty}^{+\infty} \Psi^*(x,t) \hat{E} \Psi(x,t) dx = \int_{-\infty}^{+\infty} \Psi^*(x,t) \hat{H} \Psi(x,t) dx$$

$$\bar{E} = \sum_{m,n} \int_{-\infty}^{+\infty} C_n^* \Phi_n^*(x) e^{\frac{i}{\hbar} E_n t} \hat{H} C_m \Phi_m(x) e^{-\frac{i}{\hbar} E_m t} dx$$

$$= \sum_{m,n} \int_{-\infty}^{+\infty} C_n^* \Phi_n^*(x) e^{\frac{i}{\hbar} E_n t} E_m C_m \Phi_m(x) e^{-\frac{i}{\hbar} E_m t} dx$$

$$= \sum_{m,n} C_n^* C_m E_m e^{-\frac{i}{\hbar}(E_m - E_n)t} \int_{-\infty}^{+\infty} \Phi_n^*(x) \Phi_m(x) dx$$

$$= \sum_{m,n} C_n^* C_m E_m e^{-\frac{i}{\hbar}(E_m - E_n)t} \delta_{m,n}$$

$$\boxed{\overline{E} = \sum_{n=1}^{\infty} |C_n|^2 E_n}$$

$$\int_{-\infty}^{+\infty} \Psi^*(x,t) \Psi(x,t) dx = 1 \qquad \boxed{\sum_{n=1}^{\infty} |C_n|^2 = 1}$$

$\therefore |C_n|^2$ **是测量E_n的几率**

$$\Psi(x,t)=\frac{1}{\sqrt{2}}[\Phi_1(x)e^{-\frac{i}{\hbar}E_1t}-\Phi_2(x)e^{-\frac{i}{\hbar}E_2t}]$$

∴测量 E_1 E_2 的几率各1/2

这是能量本征态的叠加态，不再是定态

*测量使系统坍缩到测量值对应的本征态

实验测到 E_1 $\Psi(x,t)\rightarrow\Phi_1(x)e^{-\frac{i}{\hbar}E_1t}$

薛定谔猫

$\Psi=\Psi_1+\Psi_2$

退相干问题一直是研究前沿

3）粒子几率

$$P=\int_0^{a/2}\Psi^*(x,t)\Psi(x,t)dx$$

$$=\frac{1}{2}\int_0^{a/2}[\Phi_1^2(x)+\Phi_2^2(x)$$

$$-\Phi_1(x)\Phi_2(x)(e^{\frac{i}{\hbar}(E_1-E_2)t}+e^{-\frac{i}{\hbar}(E_1-E_2)t})]dx$$

令 $E_2-E_1=\hbar\omega_{21}$

$$P=\frac{1}{2}\int_0^{a/2}[\Phi_1^2(x)+\Phi_2^2(x)-2\Phi_1(x)\Phi_2(x)\cos\omega_{21}t]dx$$

$$P = \frac{1}{2} - \frac{4}{3\pi}\cos\omega_{21}t$$

$$P_{\max} = \frac{1}{2} + \frac{4}{3\pi} \approx 0.9244 \qquad P_{\min} = \frac{1}{2} - \frac{4}{3\pi} \approx 0.0756$$

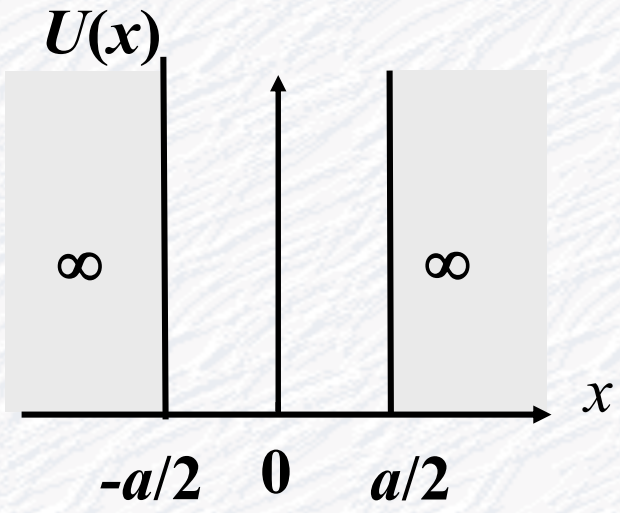
*特征时间

几率密度随时间周期变化

显然这时特征时间为 $\Delta t \sim \frac{2\pi}{\omega_{21}} = \frac{h}{\Delta E}$

- 坐标平移

$$U(x)=U(-x) \qquad \hat{H}(x)=\hat{H}(-x)$$



空间反射不变性

$$|\Phi(x)|^2=|\Phi(-x)|^2$$

$$\Phi(-x)=e^{i\delta}\Phi(x)$$

两次反射变回自身 $\rightarrow$ $\Phi(x)=(\pm 1)\Phi(-x)$

偶宇称态: $\boxed{\Phi(x)=\Phi(-x)}$

奇宇称态: $\boxed{\Phi(x)=-\Phi(-x)}$

$$\Psi_E(x,t) = C\Phi_E(x)\, e^{-\frac{i}{\hbar}Et} \qquad \hat{H}\Phi(x) = E\Phi(x)$$

- **阱内：** $$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\Phi(x) = E\Phi(x)$$

$$\Phi(x) = A\cos kx + B\sin kx$$

- **阱外：** $$\Phi(x) = 0$$

$$\Phi(\pm a/2) = 0 \quad \Rightarrow \quad A\cos ka/2 \pm B\sin ka/2 = 0$$

$$A\cos ka/2 = 0$$

$$B\sin ka/2 = 0$$

A, B 不同时为零

$$\Phi(x) = A\cos kx + B\sin kx$$

(i) A=0 奇宇称态

$$\sin ka/2 = 0 \qquad ka/2 = n\pi \to k = \frac{2n\pi}{a}$$

$$\Phi_n(x) = \sqrt{\frac{2}{a}}\sin\frac{2n\pi}{a}x \qquad (n = 1,2,3,\cdots)$$

(ii) B=0 偶宇称态

$$\cos ka/2 = 0 \qquad ka/2 = (n+\frac{1}{2})\pi \to k = \frac{(2n+1)\pi}{a}$$

$$\Phi_n(x) = \sqrt{\frac{2}{a}}\cos\frac{(2n+1)\pi}{a}x \qquad (n = 0,1,2,3,\cdots)$$

总起来

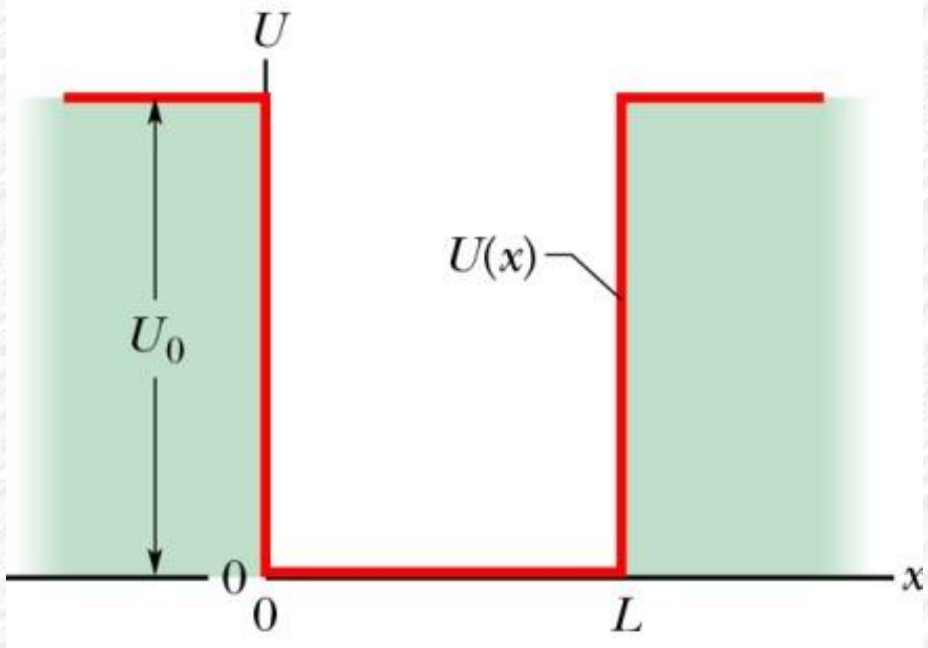
$$\Phi_n(x)=\begin{cases}\sqrt{\dfrac{2}{a}}\cos\dfrac{n\pi}{a}x & (n=1,3,5,\cdots) \quad \text{偶宇称} \\ \sqrt{\dfrac{2}{a}}\sin\dfrac{n\pi}{a}x & (n=2,4,6,\cdots) \quad \text{奇宇称}\end{cases}$$

$$E_n=\frac{\hbar^2k_n^2}{2m}=\frac{\hbar^2n^2\pi^2}{2ma^2}$$

能量最低态是偶宇称态

宇称守恒

偶宇称态和奇宇称态之间不能相互跃迁



- **有限深势阱:**

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)$$

$$\hat{H}\Phi(x) = E\Phi(x)$$

- **阱内:** $\Phi_1''(x) + k_1^2\Phi_1(x) = 0$ $\quad k_1^2 = \frac{2mE}{\hbar^2}$

- **阱外:** $E < U_0$ $\quad \Phi_2''(x) - k_2^2\Phi_2(x) = 0$

$E > U_0$ $\quad \Phi_2''(x) + k_2^2\Phi_2(x) = 0$

边界条件

- 波函数连续
- 波函数的导数连续

界面

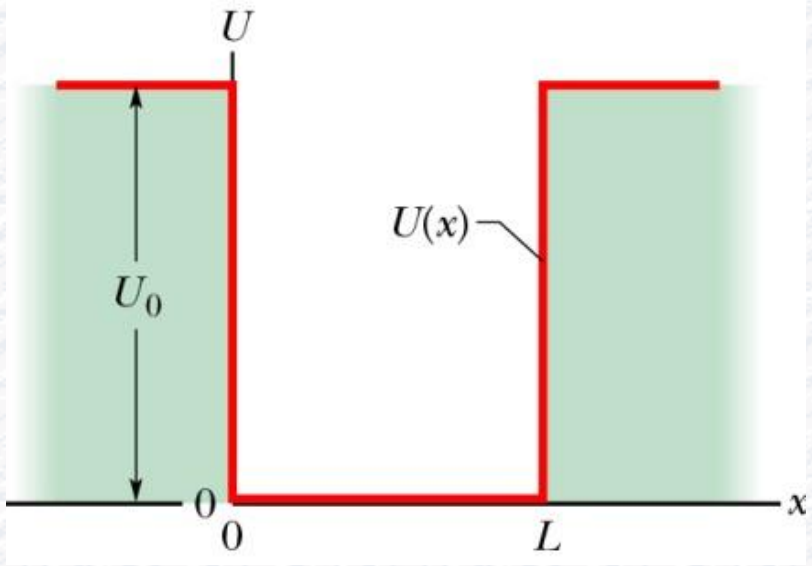
$$\Phi_1(0) = \Phi_2(0)$$

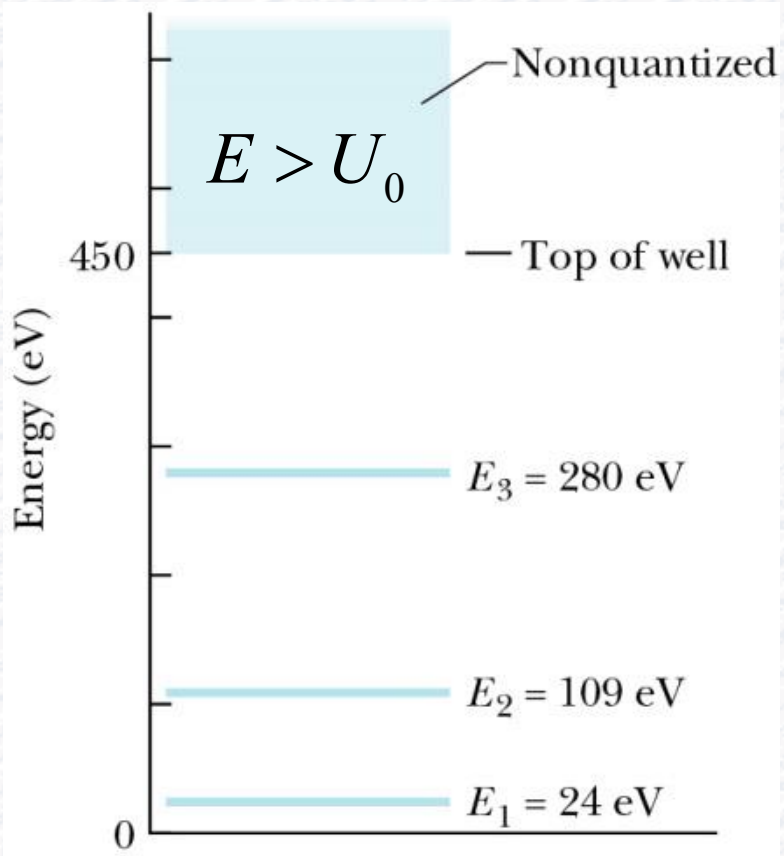
$$\Phi_1'(0) = \Phi_2'(0)$$

$$k_2^2 = \frac{2m|U_0 - E|}{\hbar^2}$$

电子 $L = 100\text{pm}$

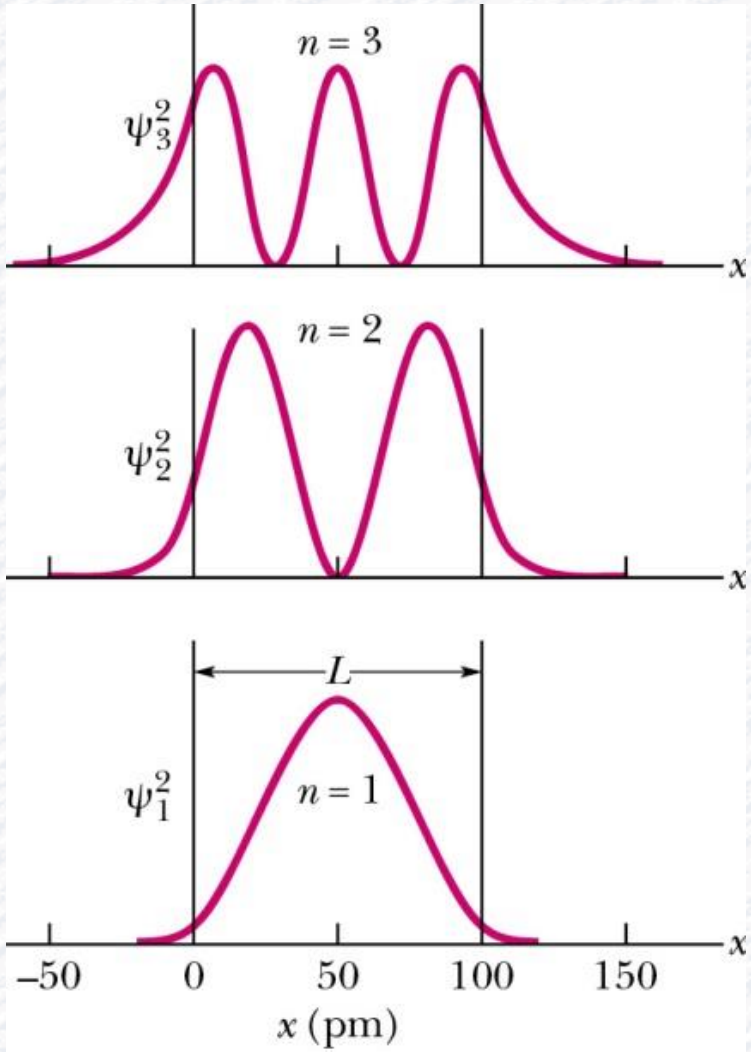
$U_0 = 450\text{eV}$

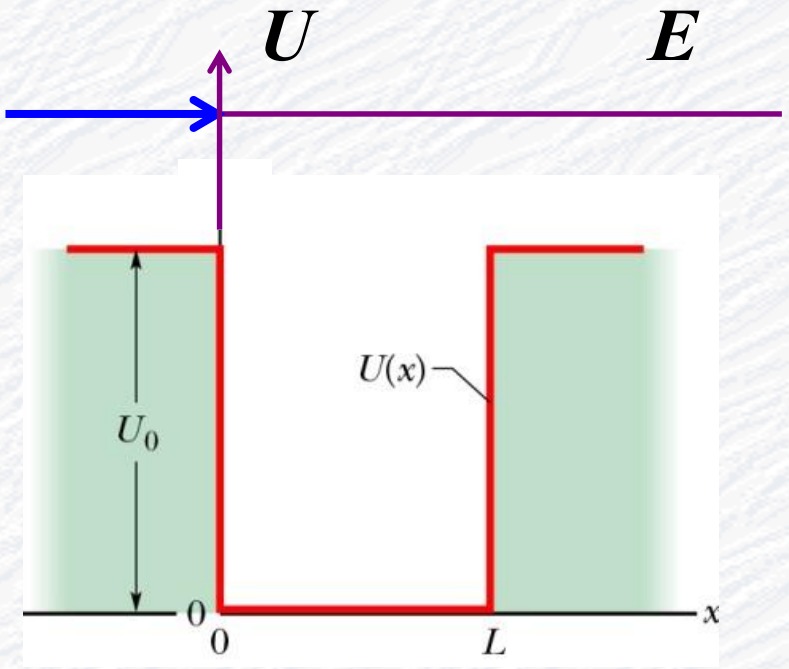




$E < U_0$

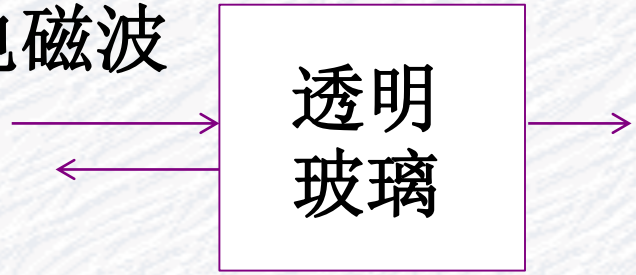
束缚态

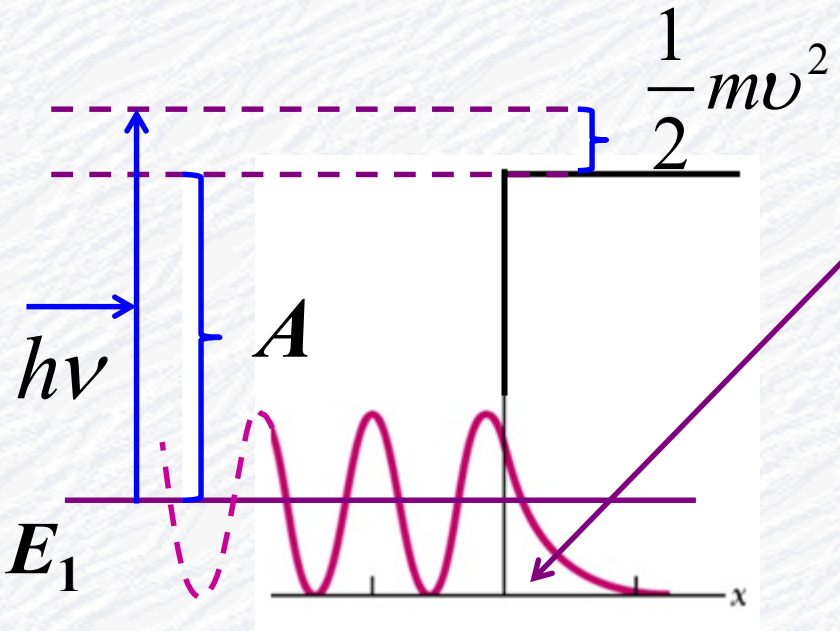




$E > U_0$ 类比

平面电磁波





- 光电效应

粒子负动能?

波粒二象性，动量不确定

衰减波(evanescent wave)

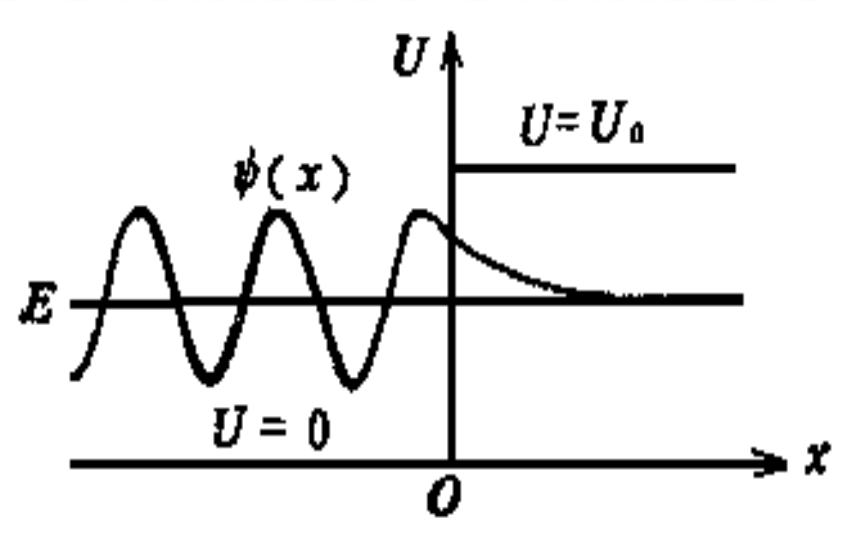
- 电子逸出金属表面的模型

§3 势垒穿透

(隧道效应)

1. 梯形势

$$U(x)=\begin{cases}0, & x<0\\ U_0, & x>0\end{cases}$$



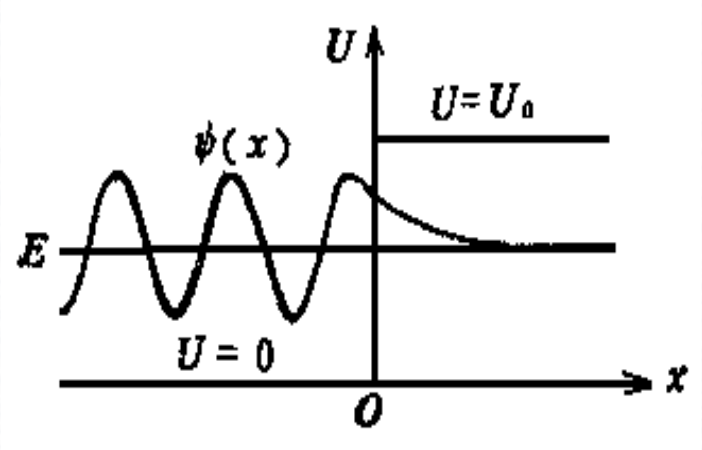
哈密顿量： $\hat{H}=-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+U(x)$

波函数： $\Psi_E(x,t)=C\Phi_E(x)\,e^{-\frac{i}{\hbar}Et}$

定态薛定谔方程： $\hat{H}\Phi(x)=E\Phi(x)$

$x<0:$ $$\Phi_1''(x)+k_1^2\Phi_1(x)=0$$

$$k_1^2=\frac{2mE}{\hbar^2}$$



$x>0:$ $$\Phi_2''(x)-k_2^2\Phi_2(x)=0$$

$$k_2^2=\frac{2m(U_0-E)}{\hbar^2}$$

通解: $$\Phi_1(x)=Ae^{ik_1x}+Be^{-ik_1x}$$

$$\Phi_2(x)=Ce^{k_2x}+De^{-k_2x}$$

波函数有限大 $\longrightarrow C=0$

简写成 $\Phi_1(x)=Ae^{ik_1x}+Be^{-ik_1x}$ （$E>U=0$, 波动解）

$\Phi_2(x)=De^{-k_2x}$ （$E<U=U_0$, 衰减解）

界面 $\Phi_1(0)=\Phi_2(0)$ $A+B=D$

$\Phi_1'(0)=\Phi_2'(0)$ $ik_1A-ik_1B=k_2D$

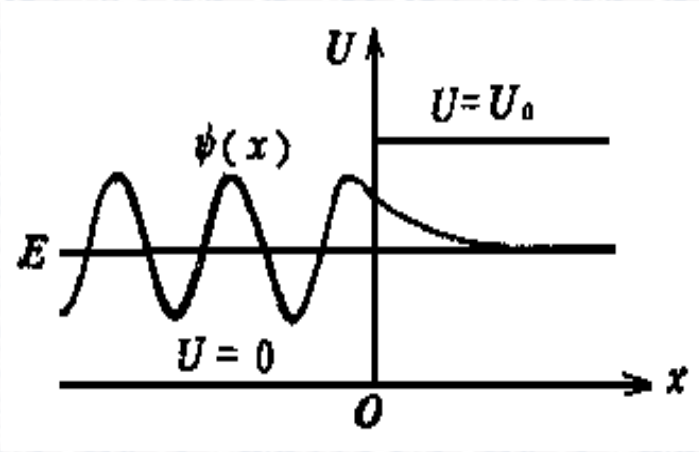
设入射波振幅 $A=1$ $B=\dfrac{ik_1-k_2}{ik_1+k_2}$ $D=\dfrac{2ik_1}{ik_1+k_2}$

$$\Phi_1(x)=e^{ik_1x}+\left(\frac{ik_1-k_2}{ik_1+k_2}\right)e^{-ik_1x}$$ （$E>U=0$, 波动解）

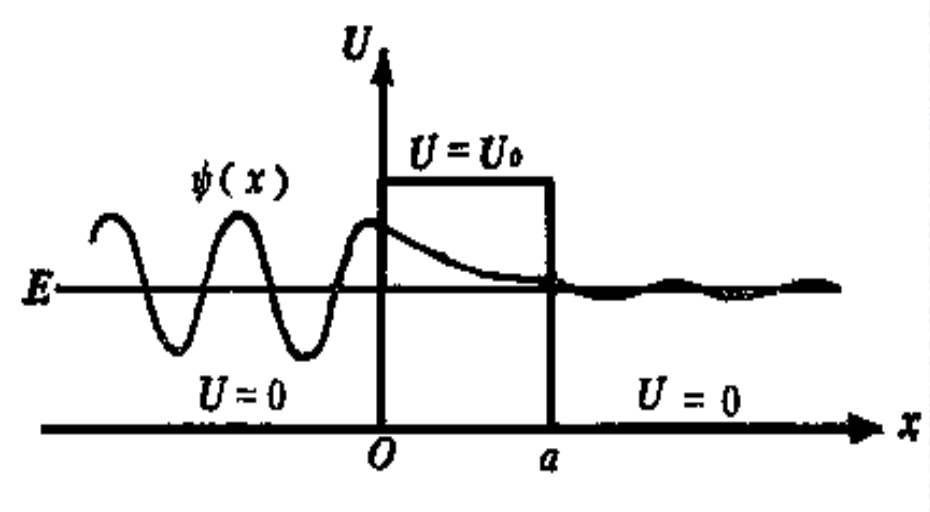
$$\Phi_2(x)=\frac{2ik_1}{ik_1+k_2}e^{-k_2x}$$ （$E<U=U_0$, 衰减解）

$$|B|^2 = \left|\frac{ik_1 - k_2}{ik_1 + k_2}\right|^2 = 1$$

界面上入射波全部反射



2. 隧道效应（势垒贯穿）



$$\Phi_1(x) = e^{ik_1x} + Ae^{-ik_1x}$$

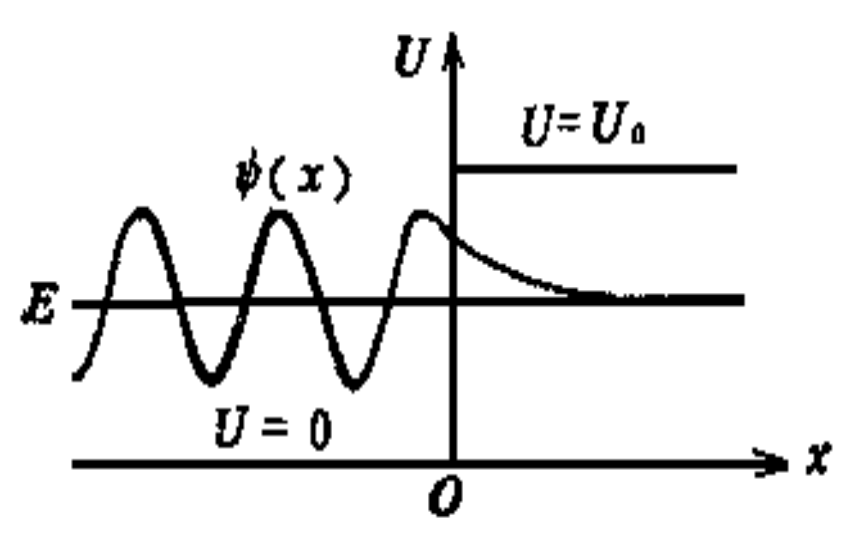
$$\Phi_2(x) = Ce^{k_2x} + De^{-k_2x}$$

$$\Phi_3(x) = Be^{ik_1x}$$

经典粒子不能穿过

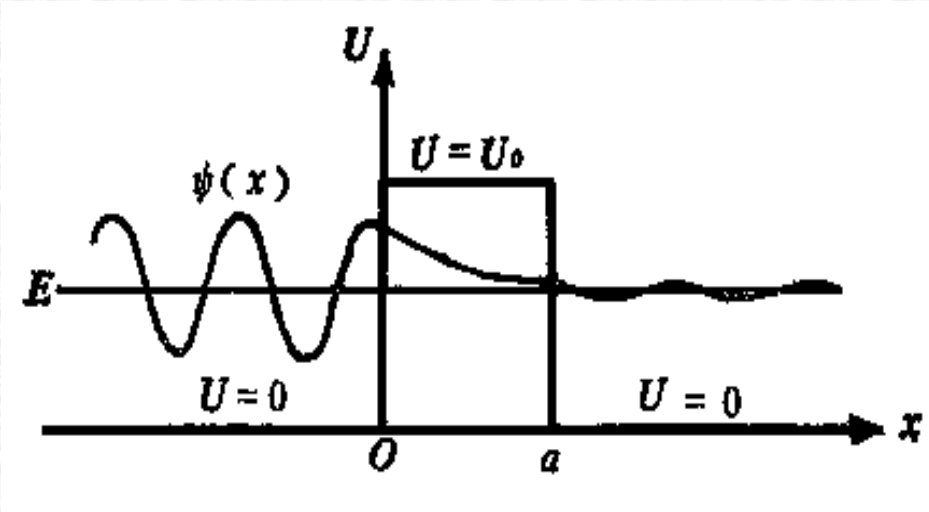
$$R = |A|^2, \quad T = |B|^2$$

$$\boxed{R+T=1}$$



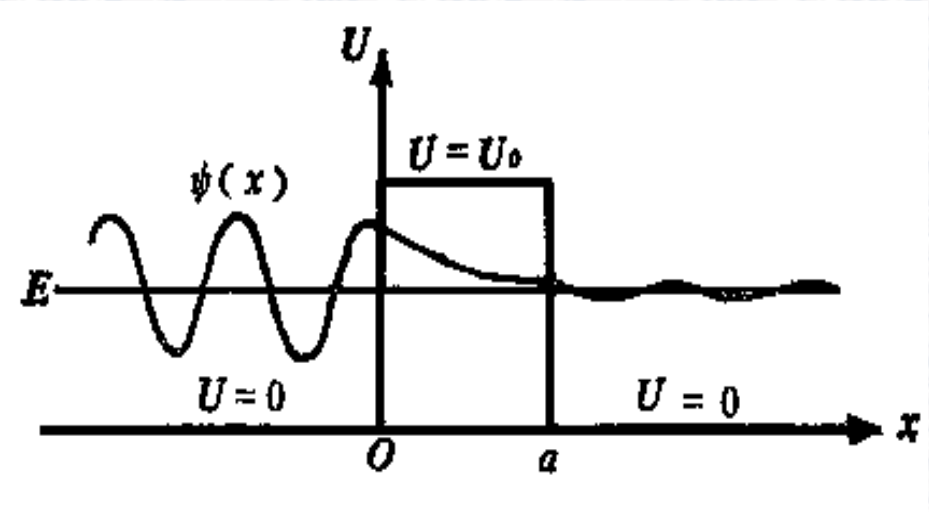
$$\Phi_2(x)=De^{-k_2x}$$

势垒既高又宽



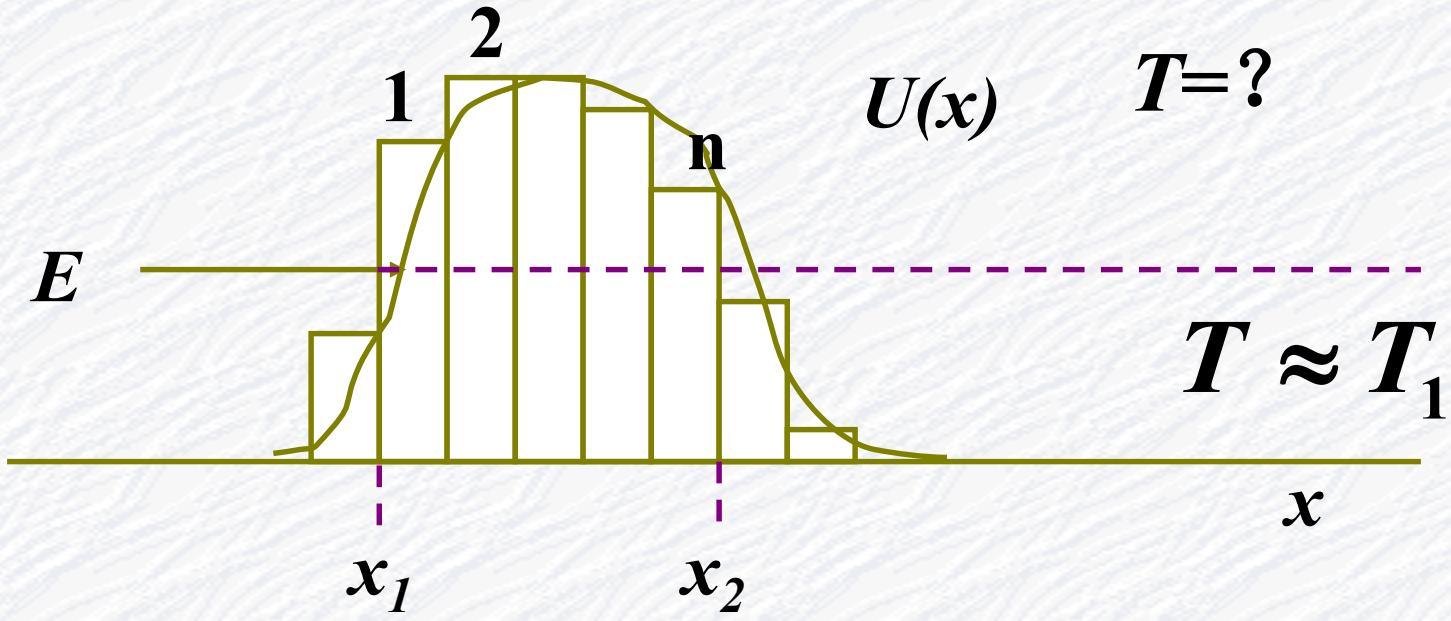
$$\frac{\Phi_2(a)}{\Phi_2(0)}\sim e^{-k_2a}$$

$$k_2^2=\frac{2m(U_0-E)}{\hbar^2}$$



$$T \sim \left|\frac{\Phi_2(a)}{\Phi_2(0)}\right|^2 \sim e^{-2k_2 a}$$

$$\boxed{T \approx e^{-\frac{2a}{\hbar}\sqrt{2m(U_0 - E)}}}$$



$T = ?$

$$T \approx T_1 T_2 \cdots T_n$$

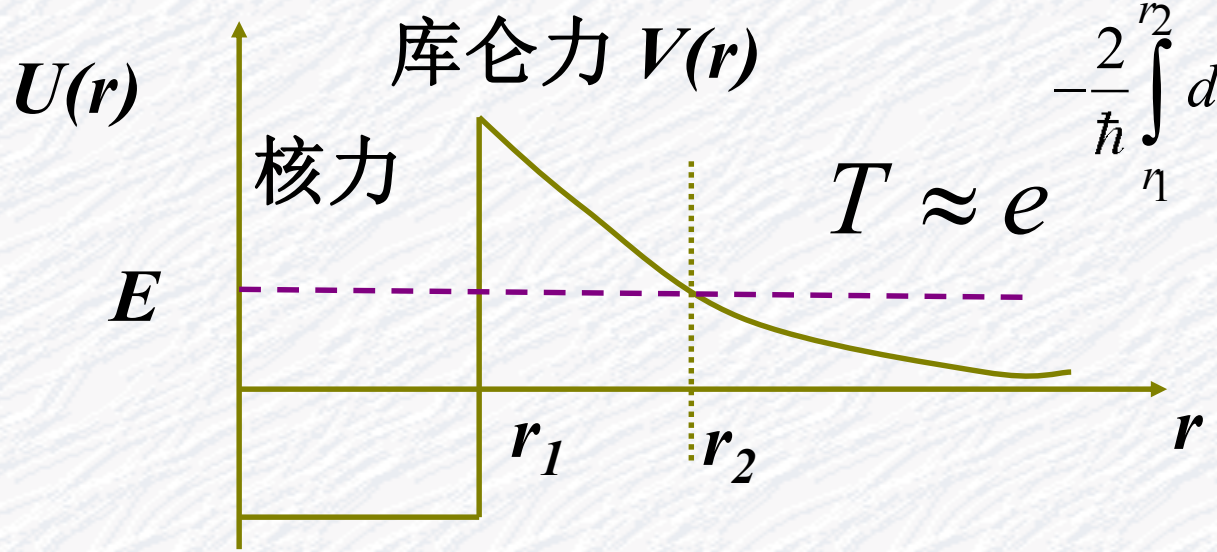
$$T_i \approx e^{-\frac{2a}{\hbar}\sqrt{2m(U_0-E)}}$$

$$a \to dx$$

$$U_0 \to U(x)$$

$$T \approx T_1 T_2 \cdots T_n \approx e^{-\frac{2}{\hbar}\int_{x_1}^{x_2} dx\sqrt{2m(U(x)-E)}}$$

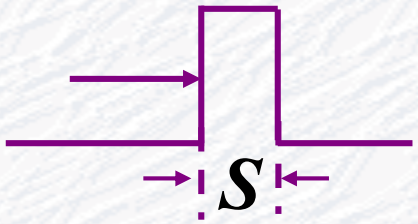
α粒子衰变几率

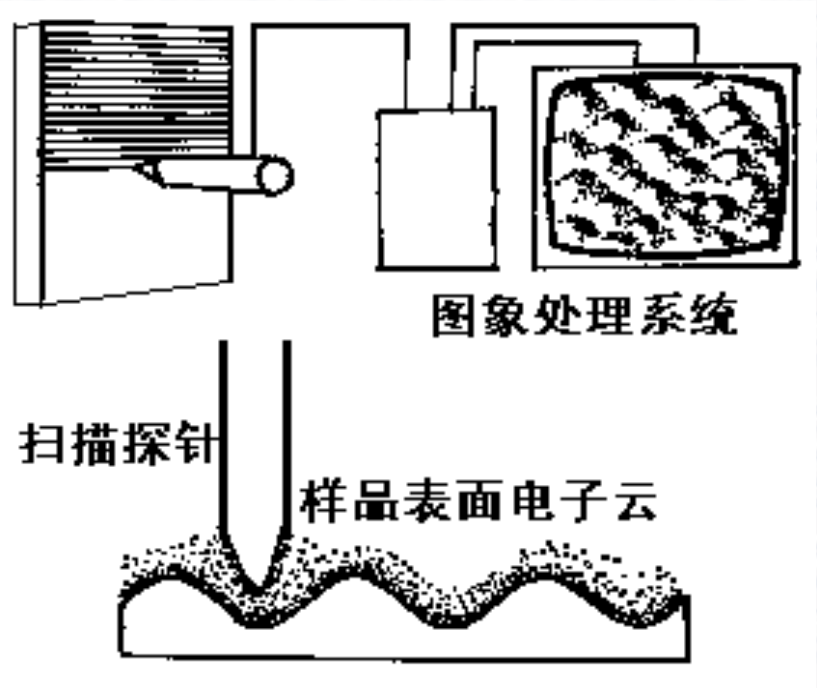


$$T \approx e^{-\frac{2}{\hbar}\int_{r_1}^{r_2} dr\sqrt{2m(\frac{2Ze^2}{4\pi\varepsilon_0 r}-E)}}$$

伽莫夫

三. 扫描隧道显微镜

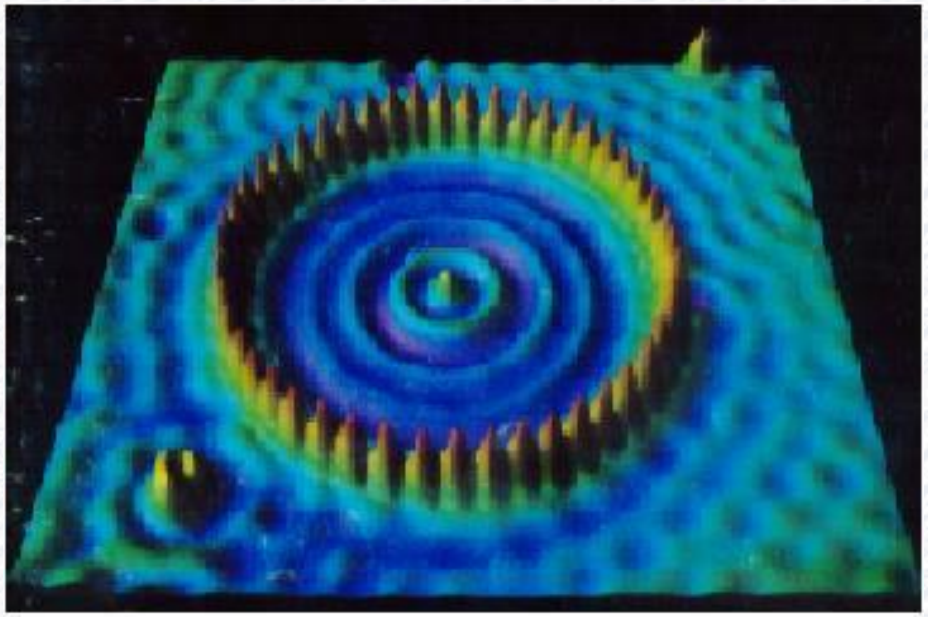




隧道电流I与样品和针尖间距离S的关系

$$I \propto Ue^{-A\sqrt{\Phi}S}$$

48个Fe原子形成“电子围栏”，围栏中的电子形成驻波.

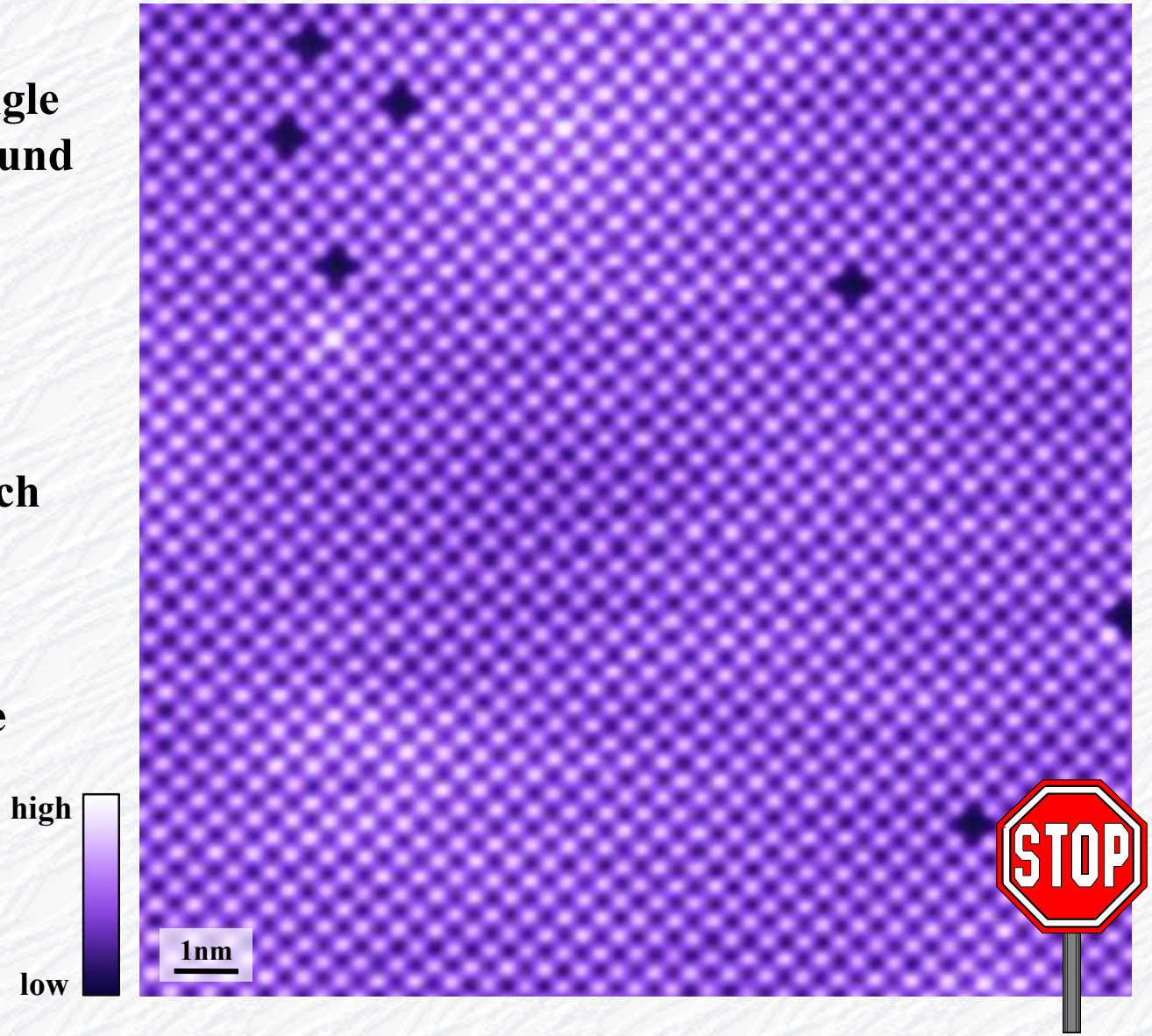


Surface topography of $Ca_2CuO_2Cl_2$

STM topography on cleaved $Ca_2CuO_2Cl_2$ single crystal, a parent compound of high temperature superconductor, with a field of view of 155Å.

The square lattice is formed by Cl atoms (each white dot is a Cl atom).

The seven dark, cross-shaped dots are surface defects. They each corresponds to a missing Cl atom.



§4 一维谐振子

一. 势函数

$$U(x)=\frac{1}{2}kx^2=\frac{1}{2}m\omega^2x^2$$

M—振子质量，ω—固有频率，x—位移

二. 哈密顿量

$$\hat{H}=-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\frac{1}{2}m\omega^2x^2$$

三. 薛定谔方程

$$\Psi_E(x,t)=C\Phi_E(x)e^{-\frac{i}{\hbar}Et}$$

定态薛定谔方程

$$\hat{H}\Phi(x)=E\Phi(x)$$

$$\Phi''(x)+\frac{2m}{\hbar^2}(E-\frac{1}{2}m\omega^2x^2)\Phi(x)=0$$

1. 能量本征值

满足量子化条件, 才存在有意义的解

$$E_n=(n+\frac{1}{2})\hbar\omega=(n+\frac{1}{2})h\nu \qquad (n=0,1,2,\cdots)$$

- 能量量子化
- 能量间隔 $h\nu$
- 最低能量(零点能)$E_0=\frac{1}{2}\hbar\omega>0$

经典振子能量

质量1g　摆长1m

$$\omega = \sqrt{\frac{g}{l}} \approx 3.3\text{Hz}$$

角度 3° $\quad U \approx mgl(1-\cos\theta) \square 10^{-5} J$

$$(n+\frac{1}{2})\hbar\omega \square U \quad \rightarrow \quad n \sim 10^{29}$$

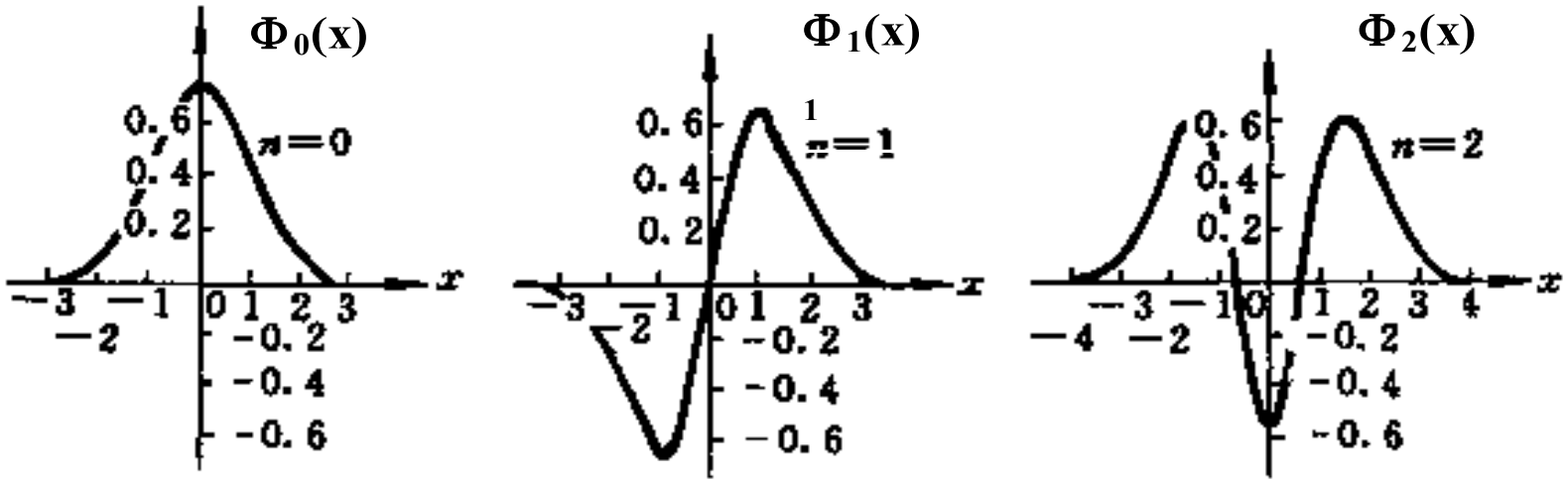
能量连续变化

2. 本征函数系的正交归一性

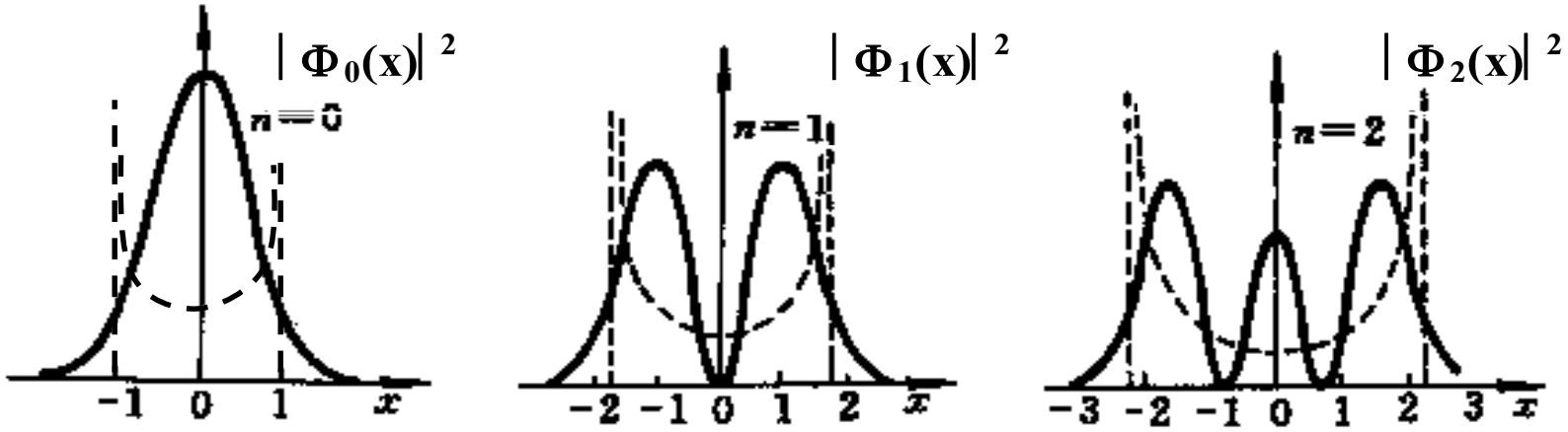
$$\int_{-\infty}^{+\infty} \Phi_m^*(x)\Phi_n(x)dx = \delta_{m,n}$$

$$\Psi(x,t) = \sum_{n=1}^{\infty} C_n \Phi_n(x) e^{-\frac{i}{\hbar}E_n t}$$

3. 本征函数和几率密度



线性谐振子的波函数



线性谐振子的位置概率密度分布

四. 与经典谐振子的比较

1. 位置几率分布

- 量子：在各处几率震荡
- 经典：在x=0处几率最小

2. 符合玻尔对应原理

$n \to \infty$

- 量子几率分布　经典几率分布
- 能量量子化　能量取连续值